MINING DEALER BEST PRACTICE

HMS Mobile Elevated Workshop

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

Due to the large size of hydraulic mining shovels, technicians are forced to complete repairs while working at heights. To mitigate this hazard, work platforms can be implemented; however, they require a lot of manual handling before/after the job. A quicker alternative is to bring in a mobile lifting work platform.

This Best Practice describes the introduction of a mobile workshop that can be raised to the working level for technicians to freely walk back and forth between the HMS and the workshop with safety barriers in place. Tools and parts can also be raised with the mobile workshop safely and technicians can move back and forth to the elevated workshop reasonably quickly (faster than climbing back and forth to ground level).

2.0 Best Practice Description

The mobile elevated workshop described in this document was inspired by the airline industry – Elevated workshops are used to service aircrafts parked at their gates. This solution is used to safely and efficiently transport goods such as: food, drinks, and luggage.

With that in mind, a mobile workshop can be built onto a truck chassis of appropriate size and carrying capacity. A lifting device is also built in to raise and lower the workshop to the appropriate height. There are many companies that will built such a facility according to specifications or modify standard containers to suit. Standard containers are good starting points because of their standard size (8ft x 20ft and 8ft x 40ft), which is standard for a large range of trucks already in existence. Special build companies also build such facilities to the standard container width of 8ft in most instances.

When specifying for such a facility, there are some minimum requirements in relation to regulations and hygiene that must be considered. For example, local legislation must be considered – Therefore, it is important to involve a specialist company for building such a facility.

Minimum requirements to consider when developing this solution:

- Truck sized appropriately.
- Lifting mechanism must have anti-drop device as per legislation, generally of a hydraulic nature.
- Stabilizers fitted at four corners of the truck and has anti-drift devices.
- Roller door at the rear or side for parts delivery. A rising tailgate is also desirable.
- Roller door at the front for mechanics entry and exit while working.
- Side door for mechanics entry when the mobile workshop is lowered to the bottom.
- An extending floor with side safety rails to extend over the truck cab to support against the machine to be worked on. There should be a provision for this extending floor to rotate a certain amount to allow it to support against the machine snugly without unacceptable gaps.

- Generator set for electric power supply for lighting any power tools.
- Air compressor for pressurized air supply.
- Water tank and pump to deliver water to the workshop for washing.
- Workbenches with vise.
- Parts wash.
- Hand wash with soap.
- First aid kit.
- Lighting.
- Power points.
- Tools for M&R activities: PM + other
- Tools for troubleshooting, diagnostics, and performance testing.
- Cutting stand for filters & screens with method to collect small quantities of waste oil.
- Storage / transport locations for backlogs and other maintenance parts
- Deliberate considerations for the type of Maintenance & Repair (M&R) work that is expected to be performed in the mobile workshop. Depending on the M&R activities, this would influence additional tooling considerations fitted in the workshop.

Figures 1 to 8 show an existing setup already in use.

Fig 1: Front view of the mobile workshop.

Fig 2: Rear view of the mobile workshop. Note the tailgate, rear roller door and side door.

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Fig 3: Raised workshop. Note stabilizers at four corners.

Fig 4: Walkway extends out and over the truck cab. Side barrier (yellow) extends out at the same time. Floor angle is hydraulically adjusted.

Fig 5: Interior rear view of mobile workshop.

Fig 6: Interior front view of mobile workshop.

Fig 7: Generator set in the truck.

Fig 8: Mobile workshop in use.

3.0 Implementation Steps

For each mine site, their own specific process should be followed with regards to introducing a new tool. However, general implementation steps should be as follows:

a. Review the need of such a tool for the site.
b. Develop a business case to justify the expense of such a case. Make some estimates of the costs or contact some manufacturers for more accurate costs. Factor in the benefits of saving labor time and downtime.
c. Review the concept with all stakeholders and develop an initial specification list.
d. Conduct a site-specific risk assessment of using such a mobile workshop on site. Develop a plan to mitigate any discovered risks.
e. Research for a suitable supplier to provide such a mobile workshop. Request for a quote and specifications.
f. Determine if such a truck-based workshop can be registered for traveling on public roads. If possible, determine if registration is required and desired.
g. Select suitable base truck in terms of size, capacity and age.
h. Finalize the specifications and authorize supplier to build.
i. Organize delivery of the completed package to site.
j. Conduct a risk assessment on the use of the mobile workshop. Action on any additional items of concern.

k. Develop a set of instructions on the use of the mobile workshop. These should also include but not limited to:
    i. When a technician is allowed and not allowed to be inside the workshop.
    ii. Precautions required to be undertaken before raising and lowering the workshop.
    iii. Conditions on site that will render the workshop unsafe for use.
    iv. Method and documents to be used for reporting any malfunctions.
    v. Safety with accidental opening of doors when the workshop is in a raised position.
    vi. Safe working load of the workshop.
    vii. Maximum number of technicians allowed in a raised workshop at any one time.

l. Develop a plan for training users and carry out training on the utilization of the workshop. The developed instructions will form part of the training.

m. Commission the mobile workshop for work.

4.0 Benefits

Benefits should include but not limited to the following.

a. Safety in performing service and repair work. Technicians are less exposed to trips and falls when travelling up and down the stairs of the machine.
b. Cost of hiring a mobile work platform.
c. Cost of transporting the mobile work platform to and from site and then to and from the machine.
d. Mobile work platforms generally do not have side walls and a ceiling. Tools and toolboxes present a risk when placed on work platforms.
e. Additional technicians' labor in performing a job when they have to carry tools and parts up and down the machine.
f. Savings in labor when having to carry components up and down for simpler jobs such as resealing a pump or motor. These can be performed in the mobile workshop.
g. With savings in labor, there is a subsequent reduction in downtime. This will equate to more production potential.

As an example with arbitrary assumptions:

Pick 10 common jobs and develop an average. An assumption is given below.
Assume no mobile work platform is hired.

Duration of a job	= 6 hours
Number of technicians on the job	= 2
Times traveled up and down the machine per job	= 10
Times traveling up and down saved with mobile workshop	= 5
Time taken to travel up and down the machine	= 5 minutes
Savings in labor per job (man-hours)	= 5 x 5 x 2 = 50 min = 0.83 hr.
Savings in downtime of the machine per job	= 5 x 5 = 25 min = 0.42 hr.
Labor savings for 20 jobs @ $150 per hour	= 0.83 x 20 x $150 = $2,490
Operating hours per year	= 6000 hours
Utilization	= 80%
Number of major jobs per year per shovel	= 20
Savings in downtime per year per shovel	= 20 x 0.42 = 8.4 hours
With utilization of 80%, addition production	= 12 x 80% = 6.7 hours
Tonnes moved if production is 4000 tonnes per hour	= 4,000 x 6.7 = 26,880 tonnes
Assume gold mine with 1:4 strip ratio	= 26,880 ÷ 5 = 5,376 tonnes ore
Assume 10 grams of mineral per tonne of ore	= 5,376 x 10 ÷ 31 = 1,734 oz gold
Benefit of addition gold with 90% recovery @ $1200/oz	= 1,734 x 90% x $1200 = **$1.87M**

The above calculation is only used as an example to show the steps in calculating the benefits of saving any downtime. Any introduction of the mobile workshop will be subject to its own individual benefits calculation with a different set of assumptions and data. This shows how the production opportunities far outweighs the savings in labor.

5.0 Resources Required

Such a project will require a champion to develop the specifications for build. Factors to consider when developing the specifications will consist of at least the following.

- Maximum height required, determined by machines to be worked on.
- Desired size of workshop.
- Equipment to be carried in the workshop.
- Capacity of the setup.
- Requirements for road use registration.
- Size of base truck.
- Electrical appliances to be used.
- Lighting required.
- Additional external lighting for night shift.
- Quantity of water for washing.
- Interlocks for exit doors when workshop is in elevated position.
- Method of extending the platform floor over the truck cab.
- Method of extending the guard rails when platform floor is extended.
- Method of angling the platform to butt against machine.
- Location of safety items such as first aid kit, eye wash, ear plugs, etc.
- Speed of raising and lowering.
- Any required heating or cooling.
- Location of control station.

6.0 Supporting Attachments / References

None

7.0 Related Best Practices

None

8.0 Acknowledgements

Existing unit built for site.
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